Deep Learning for Social Sciences

Department of Politics and Public Administration, Konstanz University

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Tracking Political Change with Embeddings of Parliamentary Speech

Project Report

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1 Introduction

This project uses an encoder-only language model to build vector representations of German parliamentary speeches and tests what these representations correspond to politically. Since we cannot inspect the embeddings directly, we approach this by testing whether they recover structure we already know from outside the model: party membership, expert ideology scores and the positions of individual MPs who change party. To do this, we clean speeches from the Open Discourse Bundestag corpus for the years 2000 to 2021 and embed them twice, once with the off-the-shelf Jina v3 model and once with a copy fine-tuned contrastively on speaker identity, before aggregating both to the MP-year and party-year level. We then validate the two spaces against party membership and the Chapel Hill Expert Survey, establish the topic structure of the debates to separate subject matter from position and trace party- and MP-level trajectories through crises and party switches.

2 Data

To address our research question, we work with German parliamentary speeches from the Open Discourse Bundestag corpus, which covers plenary speeches from 1949 to 2021 and is openly available on the Harvard Dataverse. The raw speeches table comprises 907,644 rows, each corresponding to a recorded floor contribution. Since Bundestag protocols capture the floor proceedings rather than curated remarks, a substantial share of these entries are not full speeches, but procedural remarks or brief questions directed at individual MPs. Next to the speech text itself, the dataset provides identifiers for the speaking politician, a faction ID that links to the corresponding party name via an auxiliary table, the politician's parliamentary position and the date of the sitting.

As an additional data source, we draw on the Chapel Hill Expert Survey (CHES), which we later use as a "ground-truth" measure of party positioning. CHES is a survey of country experts that asks scholars of European politics to place national parties on a range of ideological and policy dimensions. Seven waves have been conducted to date and a set of core questions is asked in all of them, most notably a party's general left-right position, its stance on European integration and separate economic and social/cultural left-right positions; this makes the data suitable for tracking party positions over time. For Germany the survey covers 16 parties, though not every party is rated in every wave, since parties enter or leave the dataset as they gain or lose parliamentary relevance. We filter the survey to Germany and use lrgen, a 0–10 scale capturing a party's general left-right position, as our external reference measure. Alongside CHES we define a small set of reference events, which anchors the temporal trajectories we derive from the embeddings against real-world political developments: four external shocks that could have shifted German political discourse, namely the 9/11 terror attacks, the Lehman Brothers collapse and the ensuing financial crisis, the Fukushima nuclear disaster and the 2015/16 Cologne New Year's Eve events, when mass sexual assaults at the peak of the refugee inflow shifted the migration debate.

2.1 Preprocessing steps

Our main dataset consists of all Open Discourse speeches from 2000 to 2021, the period we later embed, and initially contains around 303,000 speeches. We first remove all rows corresponding to chairing turns, ministerial statements and guest speakers, as well as entries without a valid politician identifier, so that only contributions by members of parliament remain, including those without faction affiliation; this reduces the dataset to approximately 135,000 speeches. We then clean the remaining texts of protocol formatting: removing the interjection markers and the orphan dashes, undoing the soft line breaks of the original PDF layout, and collapsing the paragraph structure into a single flat line, since paragraph breaks carry no signal for the embedding model. We also trim the standard salutations that open most speeches; comparing the embeddings of the trimmed and untrimmed speeches, we decide to work with the trimmed version. To find a lower length bound, we tokenize the cleaned speeches with the Jina v3 multilingual tokenizer (XLM-R based, 8192-token context) and inspect the length distribution together with samples from each length band. Speeches shorter than 100 tokens turn out to be almost exclusively procedural remarks, brief interjections and short factual questions rather than real speeches; we therefore set 100 tokens as the minimum, which leaves us with 104,960 speeches. At the upper end, 8 speeches exceed the encoder's context window and are truncated to that limit. Finally, we attach the human-readable party abbreviation from the factions table and normalise inconsistent labels.

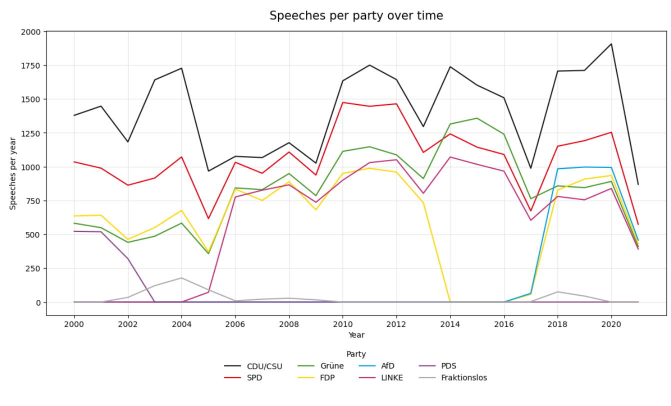
Beyond these speech-level steps, we also have to choose the unit at which speeches are aggregated for the later embedding-based trajectories. A single speech is too noisy on its own, since it is dominated by whatever topic happens to be debated on a given day. Aggregating at the MP-month level does not solve this: the median MP-month contains only 1 speech and 77% fewer than 3, so most cells would depend on very little text. So we decided to use MP-year whenever the individual MP is the object of study, where the median cell contains 6 speeches (a stable average). Whenever the party is the object of study, we use party-year, obtained by averaging the MP-year embeddings of all MPs of a party in a given year; here the median cell rests on 900 speeches and only 6% of cells fall below 50.

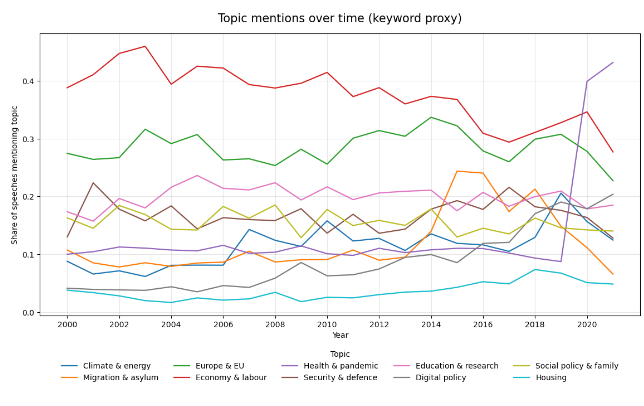
2.2 Exploratory analysis

Before embedding the speeches, we describe the cleaned corpus to get a sense of who speaks, how much and about what. The 104,960 speeches come from 1,735 unique MPs, 20 of whom appear under more than one party during the observation window and are therefore potential party switchers. The number of speakers per party follows the size of the parliamentary groups, with CDU/CSU (602 MPs) and SPD (531) contributing by far the most, ahead of FDP (198), Grüne (156) and LINKE (129). The speeches per MP run in the opposite direction: Grüne (117.6) and LINKE (104.5) MPs take the floor more than twice as often as their CDU/CSU (51.6) and SPD (43.9) colleagues, and the same gap shows up in the medians. A plausible explanation is that smaller groups have to cover the same range of policy fields with far fewer members. Speaking activity is unevenly distributed within the parties as well: an MP gives 60 speeches on average, but the most active 10% account for 37% of all speeches, with at least 139 each. Taken together, CDU/CSU and SPD contribute a little over half of the corpus. The remaining parties contribute far less, for reasons that lie in their parliamentary presence rather than their speaking behaviour: the AfD (3,496) entered the Bundestag in 2017 and the PDS (1,361) merged into LINKE in 2007; MPs without faction affiliation account for 619 speeches. Speech length differs as well: CDU/CSU and SPD speeches are the longest (median around 1,005 tokens), while Grüne, LINKE, FDP and AfD speeches are around 200 tokens shorter (medians between 767 and 822).

We further compute the type-token ratio (TTR) per party to see how varied each party's vocabulary is. The raw values differ considerably, from 0.013 for CDU/CSU to 0.096 for MPs without faction affiliation, but this ordering follows the amount of text per party: the more speeches a party contributes, the more often the same words reappear. We therefore recompute the TTR on samples of equal size (248,515 word tokens per party). The differences then disappear, with all parties between 0.10 and 0.11 and the AfD at the top with 0.115, so we find no meaningful differences in vocabulary between the parties. The cleaned corpus finally contains no missing values in any of the variables we use and no empty speeches, a direct result of the filtering: in the raw dataset, 4.2% of the rows carry no valid politician identifier and 58.8% no valid faction identifier, and all of these are removed.

Speaking activity develops largely in parallel across parties: the number of speeches drops in the federal election years when the Bundestag sits less often. The differences in level between parties stay the same over the whole period and reflect the size of the parliamentary groups, with CDU/CSU on top, followed by SPD. Parties that are not present in every legislative period behave differently: the PDS disappears after 2005 and is replaced by LINKE from 2006, the FDP drops to zero between 2013 and 2017, and the AfD only appears from 2017. The number of MP-year cells per party therefore varies over time, which we keep in mind when interpreting party-level trajectories.





Turning to the topics mentioned in the speeches, which we approximate with a keyword list rather than the BERTopic model used later, two groups emerge. The first is present throughout: economy and labour is mentioned in 28–46% of the speeches in every year and remains the most frequent topic, followed by Europe and the EU (26–34%), education and research, and social policy and family, which all stay at a similar level. The second group is event-driven: migration and asylum stays around 9% until 2014 and jumps to roughly 24% in 2015 and 2016 before falling back, health and pandemic remains close to 10% for two decades and then rises to about 40% in 2020 and 43% in 2021, and security and defence peaks in 2001. Digital policy follows a third pattern, a steady increase from around 4% in 2000 to around 20% in 2021 that cannot be traced back to a single event.

3 Embedding Model

Having cleaned the corpus and settled on MP-year and party-year as our units of analysis, we next turn to how the underlying speech-level embeddings are built. We compare an off-the-shelf sentence embedding model to a version we fine-tune ourselves, since a pretrained model has not been adapted specifically to parliamentary language and we want to check whether adapting it to this domain sharpens the political signal in the resulting representations.

3.1 Fine-Tuning and Embedding Corpus Split

Since fine-tuning contrasts multiple speeches from the same speaker, we restrict the corpus to speakers with at least 20 speeches (1,216 MPs across the eight parties) and sample roughly 25% of each speaker's speeches for the fine-tuning subsample, bounded between 8 and 40 per speaker to prevent high-volume speakers from dominating while keeping most of each speaker's speeches (on average 74.4%) available for the embedding corpus. This yields 22,513 speeches (21.4% of the corpus). From the remainder we draw a 150-per-party, stratified party-probe set of 1,200 speeches, split into 720 for checkpoint selection and a held-out 480 for the final comparison; both remain part of the 82,447-speech embedding corpus used for the main analysis, disjoint only from the fine-tuning subsample.

3.2 Off-the-Shelf Baseline

As a baseline we encode the full 82,447-speech embedding corpus with the pretrained Jina v3 embedding model (jinaai/jina-embeddings-v3-hf). This model was chosen because it is multilingual and therefore covers German, and because its 8,192-token context window is long enough to encode almost every speech in full rather than in truncated form. For pooling, we take the mean over token representations and then apply L2 normalization. At this stage no fine-tuning is applied, which both gives us a reference representation for the downstream MP-trajectory analysis and lets us isolate what fine-tuning later adds on top of it. We factor the pooling and batched-encoding logic into a single shared module used by both the baseline and fine-tuned embedding runs. This ensures that the two embedding variants use the same pooling and normalization procedure, so differences in the resulting representations are not attributable to differences in the embedding post-processing. The one encoding parameter we vary across stages is the maximum sequence length: we use a training input truncation length of 256 tokens for fine-tuning batches for speed, 512 tokens for probe evaluation, and Jina v3's full 8,192-token context for the final full-corpus encoding pass, so that speeches in the final embedding corpus are embedded at their full context length rather than at the truncated length used during fine-tuning.

3.3 Contrastive Fine-Tuning

To adapt the embedding space to the specific structure of parliamentary speech, we fine-tune the Jina v3 backbone using Low-Rank Adaptation (LoRA, rank 8, applied to the attention projection matrices), keeping the base model weights frozen and training only 0.28% of its parameters (1.57M of 560.9M). Rather than fine-tuning on party labels, which would optimize for party separability without necessarily reflecting meaningful semantic structure, we use a speaker-identity supervised contrastive loss. Speeches from the same speaker within a batch are treated as positive pairs, and speeches from all other speakers as negatives. The intuition is that an individual politician's speeches should be more similar to one another in embedding space than to other speakers', which should indirectly sharpen the party signal to the extent that party is itself a meaningful axis of variation in how MPs speak.

We construct batches with a speaker-grouped (PK) sampler (P = 8 speakers, K = up to 4 speeches per speaker) rather than random shuffling, since random batches would rarely contain more than one speech per speaker and could not supply positive pairs. We train for 3 epochs (learning rate 2e-5, temperature 0.05) and evaluate the party-probe validation set every 50 steps, retaining only checkpoints that improve on probe accuracy and merging the best-performing LoRA weights back into the base model at the end of training. Over the course of training, validation accuracy on this set rises from a step-0 baseline of 0.315 to a peak of 0.389. We do not report this peak as our headline fine-tuning result, since it reflects the best of many repeated evaluations against the same 720-speech validation set used for checkpoint selection and is therefore optimistically biased. The held-out comparison, on the 480-speech test set that plays no role in checkpoint selection, is reported in Section 3.4 and is noticeably lower.

3.4 Party-Probe Evaluation

We compare the off-the-shelf and fine-tuned embeddings on the party-probe test set of 480 speeches, which is excluded from fine-tuning and checkpoint selection but remains part of the broader embedding corpus. For this comparison we re-embed these speeches in a separate pass, running a freshly loaded off-the-shelf model and the fine-tuned model one after the other, so that the two sets of embeddings differ only in the model that produced them. We measure party separability with 5-fold cross-validated logistic regression accuracy and with silhouette score against the eight party labels.

Model	Probe accuracy	Silhouette
Off-the-shelf	0.306	−0.0074
Contrastive fine-tuned	0.340	−0.0117

Fine-tuning improves party classification accuracy from 30.6% to 34.0%. Both models are well above the 12.5% chance level for eight balanced classes, but the improvement from fine-tuning itself is small. This suggests that the speaker-contrastive objective, despite never seeing party labels directly, does sharpen some party-correlated structure in the embedding space, plausibly because individual speaker style is itself correlated with party affiliation. Note that these numbers are not directly comparable to the party classification results reported later in Section 4.1: here we classify individual speeches on the 480-speech probe test set as a held-out evaluation of the selected model, whereas the later analysis classifies aggregated embeddings across the full corpus, which is both a different unit of analysis and a different, larger evaluation set.

The silhouette score becomes more negative under fine-tuning ($-0.0074 \rightarrow -0.0117$), indicating that the fine-tuned representations do not exhibit tighter or more distinct party clustering according to this metric, despite the improvement in linear classification accuracy. We see this as a meaningful discrepancy worth flagging rather than smoothing over: probe accuracy and silhouette capture different notions of separability, since a linear classifier can exploit a direction along which classes are pulled apart even if the classes remain diffuse, unevenly sized, or non-convex, which is plausible here given both the class-imbalanced fine-tuning data and the fact that we optimize the model for speaker identity, not party. The discrepancy also foreshadows a broader tension that returns in Sections 4.1 and 4.2 below: the clustering-oriented validation checks benefit from a separated embedding space, while the ideology-alignment checks are closer to a correlation task, where what matters is a coherent left-right axis rather than compact clusters, and a single fine-tuning objective is unlikely to optimize both at once. We then use the fine-tuned backbone to re-embed the full 82,447-speech embedding corpus, producing the second embedding set that we aggregate to the MP-year and party-year level and validate against party membership and the CHES scores in the next section.

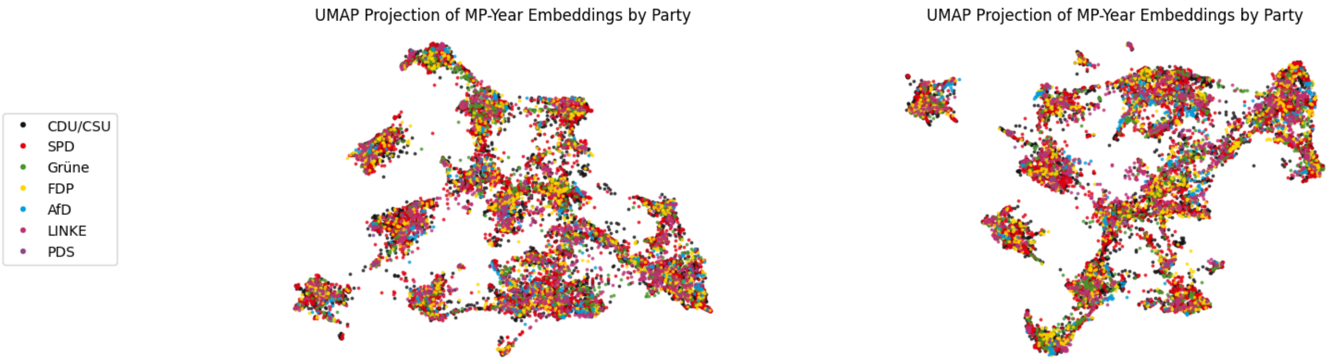
4 Test Embeddings

Before using the speech embeddings for our trajectory and event analysis, we test whether they capture politically meaningful structure at all, comparing off-the-shelf to fine-tuned embeddings. Since a 1024-dimensional vector has no inherently correct meaning, we cannot judge embedding quality directly; instead we test whether the embedding geometry aligns with independent, externally known political structures, specifically whether (1) party membership is recoverable from the embeddings and (2) the geometry aligns with an external ideology measure (from CHES).

4.1 Party Affiliation Recovery

To test whether party affiliation is recoverable, we first project the aggregated MP-year embeddings to two dimensions using UMAP for a visual check, then test on the full-dimensional embeddings via KMeans clustering and logistic regression classifiers. Neither UMAP projection shows visually separable party clusters. This could reflect several factors, e.g., topical or stylistic variation dominating over ideological signal, or simply information loss from reducing 1024 dimensions to two.

We perform KMeans clustering on the full-dimensional embeddings and evaluate it using purity and the Adjusted Rand Index (ARI). Purity captures the share of embeddings assigned to the majority class within their respective cluster; however, with imbalanced datasets this measure can be misleading, as it is easy for most or all clusters to end up dominated by the same, largest class. Indeed, for both sets of embeddings all seven clusters are assigned the majority class (CDU/CSU), so purity (0.375) simply reflects the CDU/CSU's share of the data rather than genuine separation. We therefore complement it with the ARI, which considers all pairs of data points and checks whether the clustering and the true party labels agree on whether each pair belongs together, corrected for the level of agreement expected by chance; our ARIs of −0.0006 (off-the-shelf) and −0.0014 (fine-tuned) are essentially at chance level.

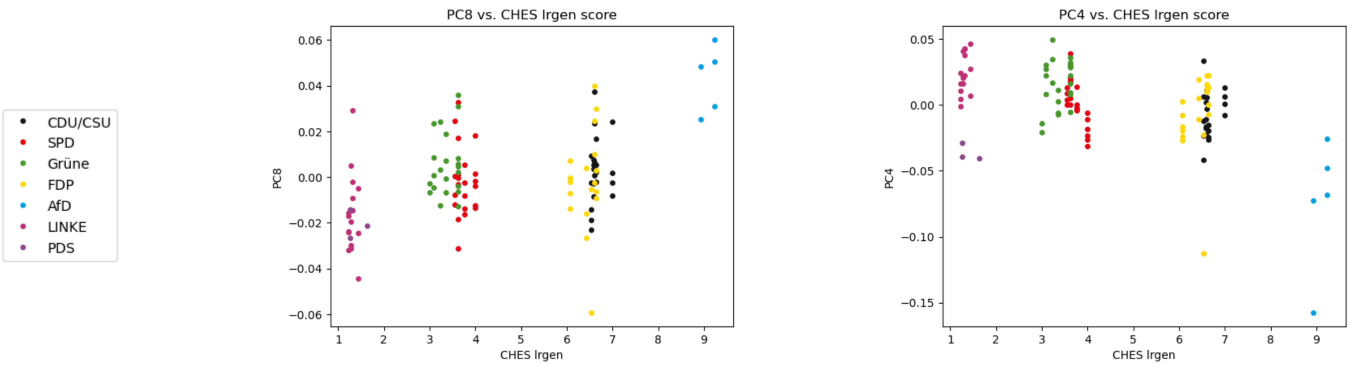


The logistic regression classifiers, in contrast, show a clear association between embeddings and party membership. Accuracy reaches 0.480 (off-the-shelf) and 0.583 (fine-tuned), both clearly above the majority-class baseline of 0.383; macro-F1 reaches 0.462 and 0.543 respectively, versus a baseline of 0.078. Since party membership was never given to the classifier and must be recovered from the raw text via the embeddings alone, this above-chance performance indicates that the embeddings capture politically meaningful structure rather than purely topical or stylistic variation and that fine-tuning further improves this.

4.2 Alignment with External Ideology Measures

We next test whether the embedding geometry aligns with the parties' general left-right position (lrgen) in CHES, first via principal component correlations, then via regressors predicting lrgen from party-year aggregated embeddings.

For both embedding sets, PC1 correlates only weakly and insignificantly with lrgen (off-the-shelf: $r = -0.173$, $p = 0.072$; fine-tuned: $r = 0.096$, $p = 0.32$). We therefore extend the analysis to the first ten components, since the ideological signal may sit in a less dominant direction. For the off-the-shelf embeddings, PC8 shows the strongest relationship ($r = 0.435$, $p < 0.001$) despite explaining only 3.2% of total variance. The plot of PC8 against lrgen is consistent with this moderate correlation: centrist parties overlap substantially, but as a group are separated from the extremes (the LINKE/PDS at one end, the AfD at the other). For the fine-tuned embeddings, PC4 is strongest ($r = -0.498$, $p < 0.001$, 7.7% of variance). The plot shows the LINKE and the AfD clearly separated from the other parties while those other parties show only weak internal ordering. The PDS does not fit the left-right pattern, scoring similarly to the AfD on PC4 despite its left-wing position.



Given this mixed evidence, we turn to a supervised regressor as a more direct test, using the same validity-test logic as for the party classifier: if lrgen is predictable well above baseline, this indicates that the embeddings encode some information about ideology. Since the number of features (1024) far exceeds the number of observations, we use Ridge regression rather than OLS, since its regularization avoids severe overfitting. For the off-the-shelf embeddings, the MSE (9.44) is slightly worse than the naive baseline (9.36) and R^2 is strongly negative (−5.38), though the correlation between predicted and true values is moderate and correctly signed ($r = 0.314$, $p = 0.054$, narrowly missing significance). The fine-tuned embeddings show the same pattern more strongly: MSE (9.23) and R^2 (−5.24) remain similarly poor, but the correlation is now moderate-to-strong and significant ($r = 0.459$, $p = 0.004$), suggesting that fine-tuning strengthens the still-incomplete ideological signal. As a robustness check, we also fit a Principal Component Regression (PCR), using the first 15 principal components as predictors for an ordinary linear regression. Its metrics are broadly consistent with those of the Ridge regression, but the correlations are weaker and statistically insignificant.

Overall, although effect sizes remain moderate and unsupervised clustering fails, the classification results and lrgen regression correlations suggest that the embeddings encode politically meaningful structure at least to some extent, but are also heavily influenced by other sources of variation such as topic and speaking style. As fine-tuning yields better results, we rely on the fine-tuned embeddings for the following analysis.

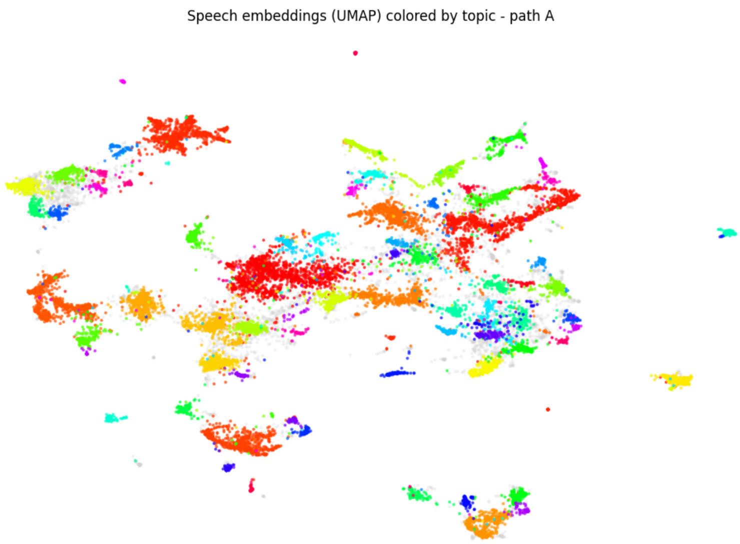
5 Topics and Party Movement Around Crises

The validation in the previous section shows that party membership is recoverable from the embeddings while the ideological signal is weak and inconsistent. Before tracing movement over time we therefore need to know what else the space is organised by, since a trajectory that looks political may be nothing more than a change of subject. We first derive the topic structure of the debates and then ask whether the events appear as movement inside those topics. All results in this section use the fine-tuned embeddings aggregated to the party-year level, and MPs without faction affiliation are excluded. We reproduce only a selection of the figures; the complete set, for every event and both topic assignments, is in the accompanying notebooks.

5.1 Topic Structure of the Debates

We run BERTopic directly on the fine-tuned speech vectors rather than letting it embed the texts again, so the clusters live in exactly the space we validated. UMAP reduces the 1,024 dimensions to five before clustering, and we require at least 150 speeches per topic, which yields 72 topics. BERTopic leaves the vague or mixed speeches unassigned, 24% of the corpus in our case; we keep this raw assignment and, in parallel, a version in which every leftover speech is reassigned to its nearest topic. Both are saved, and the analyses below are run on the raw assignment where a topic should mean a speech that genuinely belongs to it, and on the reassigned one where full coverage matters. Raising the minimum topic size above 150 produced a single dominating category, so we kept this configuration and address granularity separately in Section 5.6.

Since BERTopic's own labelling strips umlauts, we build labels from an umlaut-preserving count vectorizer with a stopword list extended by plenary boilerplate, naming each topic by its three most over-represented words. The topics are balanced, with the largest holding 6.84% of all speeches.



Projecting the speech embeddings into two dimensions and colouring them by topic shows how strongly the space is organised by subject: the topics form clean, separated islands. This is a speech-level projection and therefore not directly comparable with the one in Section 4.1, which was computed on MP-year aggregates, but it shows that subject matter is a dominant axis of variation in the vectors those aggregates are built from. The four events are represented by the ISAF-mandate, covered-bond, reactor-safety and waste-storage, and citizenship and Dublin-regulation topics; the full labels are in the notebooks.

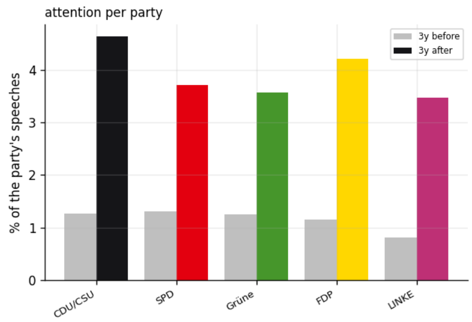
5.2 Movement Without Topics

Before filtering by topic, the party-year embeddings on their own show two things. Party trajectories in the first two principal components are tangled rather than separated, and every party ends the period closer to the others than it started. On the first component all parties rise together, from roughly −0.08 in 2000 to +0.11 in 2021, with the steepest section after 2013. A shared, parallel movement of this kind is a time effect: the component is tracking how parliamentary language changed over two decades, not who moved where.

Constructing a left-right axis directly, as the difference between LINKE/PDS and CDU/CSU centroids, orders the parties from the LINKE (−1.89) and the AfD (−0.80) at one end to the SPD (+0.75) and the CDU/CSU (+0.84) at the other, with the Grüne, the FDP and the PDS in between. Ignoring the anchor parties, the Grüne and the FDP land plausibly while the AfD and the SPD are badly misplaced. This is the horseshoe: the AfD's oppositional register resembles the LINKE's, and the long-governing SPD resembles the CDU/CSU. The axis is tracking government versus opposition and topic, not ideology, which is consistent with the mixed correlations against CHES in Section 4.2 and is the reason the event analyses below anchor their axes inside a single topic.

5.3 Does the Agenda Move?

The first question is not about position but about attention: does an event change how much is said about its topics at all? We measure the share of a year's speeches falling on the event's topics and for each party the share of its own speeches in the three years before and after.



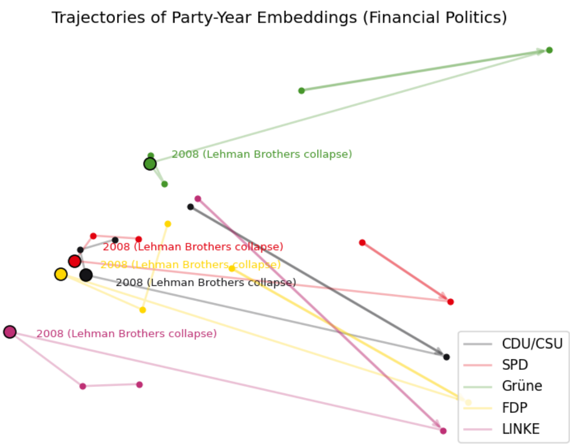
Lehman is the textbook case. Attention runs at about 1% through the early 2000s, rises to 3.4% in 2008, peaks at 5.9% in 2009 and decays back to roughly 1.8%. Every party roughly triples in the three years after against the three before, from around 1.2% to between 3.5% and 4.7%, so nobody owns the crisis; that is what a pure agenda effect looks like. Migration behaves similarly but with an owner: attention peaks in 2015, a year before the Cologne New Year's Eve that dates the event, reaches a second peak in 2018, and the AfD devotes 6.03% of its speeches to the topic over the whole period against 2.70–4.13% for the others.

The other two events do not fit the pattern. The nuclear topics peak in 2010 at 5.2%, the year before the disaster, driven by the Laufzeitverlängerung of autumn 2010, the extension of the reactors' operating lives. Attention afterwards never returns to that level; the Grüne, who own the topic, go slightly down (6.15% to 4.47%). For 9/11 the corpus records the parliamentary consequences rather than the attack: attention rises from 0.06% in 2000 to 1.2% in 2001, but the real peaks are 2007 and 2008, the years of contested ISAF mandate renewals.

5.4 Do the Parties Move?

To test whether crisis events left a visible trace in how parties spoke about the corresponding topic, we filter the speeches by topic and apply six complementary approaches to the party-year aggregated embeddings: (1) two-dimensional PCA trajectories with all parties plotted jointly, to see whether the event triggers parties to converge or drift apart from one another; (2) the same trajectories split into one plot per party, to check whether the event triggers a markedly larger movement for a given party; (3+4) one-dimensional PC1 trajectories over time (raw and demeaned by year), to make the year-to-year development easier to follow; (5) trajectories of the single most active MP per party ("specialists"), to remove the effect of a changing pool of speakers, whose differing styles can trigger shifts that reflect stylistic variation rather than a real change in position; and (6) the year-over-year cosine distance between consecutive party embeddings, which is quantitative rather than visual and takes all embedding dimensions into account rather than only the two or one retained by PCA.

Only for the financial crisis do the joint two-dimensional trajectories show a pattern consistent with an event effect: the CDU/CSU, SPD and FDP converge sharply in 2008 and the Grüne move noticeably closer to this group. For the other three events, no comparable convergence appears. This is especially surprising for Fukushima, as the governing coalition (CDU/CSU and FDP) reversed its own recent extension of reactor lifespans in direct response to the disaster, thereby adopting the position of the SPD, Grüne and LINKE.



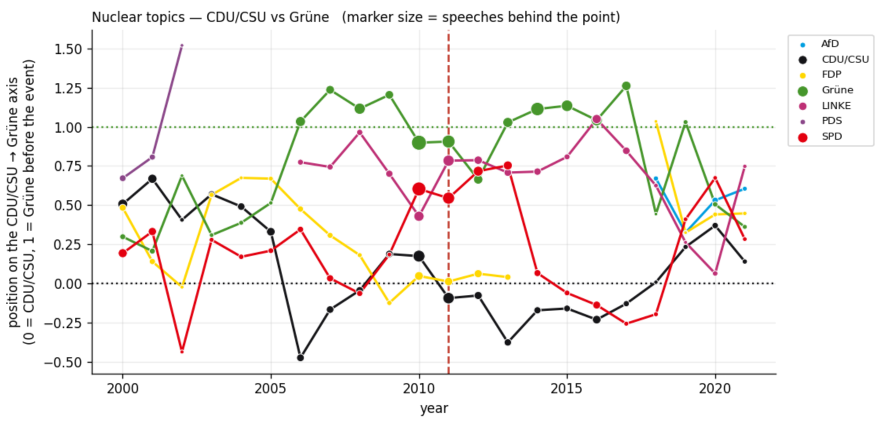
The per-party trajectories and the year-vs.-PC1 plots show the same exception: visible effects appear only for the financial crisis. For the other events, party-level trajectories move noticeably at times and not only around the crisis years, and this holds after demeaning by year as well. Kinks coincide with the events so rarely that those few overlaps are more plausibly coincidental than systematic. The specialist trajectories do not improve this picture either.

The cosine distance analysis complicates this picture further. For all three topics with sufficient data (financial politics, nuclear energy, migration), party embeddings show low year-over-year distances around the respective event, rather than an elevated distance as would be expected from a genuine shock, including for the financial crisis. Taken together, the six methods detect a signal for the financial crisis in three of the six approaches, while 9/11, Fukushima and the Cologne New Year's Eve show no consistent alignment with the corresponding crisis year across any of the methods used.

5.5 Anchored Positions

A principal component chooses its own direction and Section 5.2 showed that this direction is mostly era and government status. Where we know which two parties should sit at the ends of a scale, we can instead build the axis ourselves. Within a topic we aggregate to party-years, subtract the yearly mean, estimate the axis between two anchor parties on the years before the event so that the ruler does not move with the data, and rescale so that the anchors sit at 0 and 1 in that reference period.

Fukushima is the clearest test case available, because the answer is documented outside the corpus: after March 2011 the CDU/CSU reversed course and legislated the phase-out it had extended months earlier. If the embeddings carry policy content, the CDU/CSU should move towards the Grüne. From 2006 on the ordering is stable and plausible, with the Grüne between 0.9 and 1.27, the LINKE between 0.7 and 1.05, and the SPD, FDP and CDU/CSU at or below 0.3, so the axis does separate the parties. The hypothesis nevertheless fails. Comparing 2007–2010 with 2012–2015, the CDU/CSU moves away from the Grüne, from 0.04 to −0.20; the Grüne drop slightly (1.11 to 0.98) and the LINKE barely moves. The one party moving clearly upward is the SPD (0.19 to 0.37), and its timing gives the reason: it sits high while fighting the lifetime extension from the opposition benches and falls below zero once it enters the grand coalition in 2014. What the axis tracks is again an opposition-versus-government register. The Merkel-Wende, the government's turnaround on nuclear power, is a documented policy reversal that left almost no trace in plenary language, which we read as a limit of the method rather than a finding about the CDU/CSU.



The migration axis, anchored on the Grüne and the AfD, has the opposite problem. The AfD separates cleanly from 2017, sitting at 0.64–0.86 while every other party stays between 0.0 and 0.4, but before 2017 all parties including the Grüne is bunched at 0.2–0.4 with no stable ordering. The axis has not found a migration-policy scale on which the house can be ranked; it has found AfD language against everyone else, and by this measure the CDU/CSU is no closer to the AfD than the SPD is. The reference period also lies entirely after the event, since the AfD did not exist in parliament before it, so the early years are projections onto a ruler built from years those parties had not yet reached.

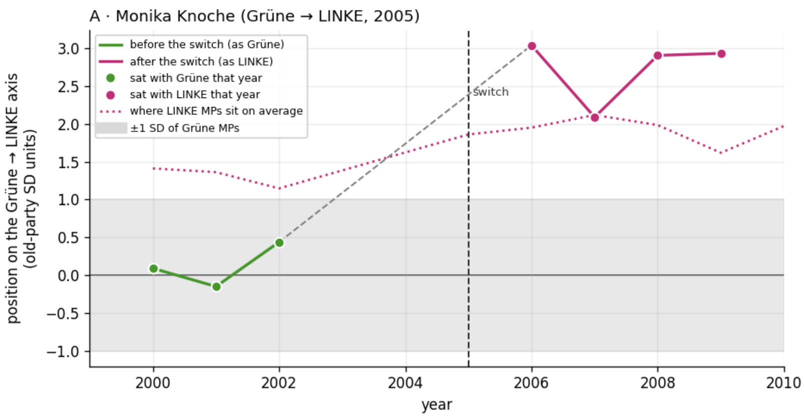
5.6 Zero Shot for Party Movement and Crisis Response

Because the results based on the BERTopic assignment were so limited, we suspected that they might partly stem from BERTopic producing topics that are narrow and not well aligned with the crises under study. We therefore tried an alternative topic assignment: we derived a set of 21 policy fields from the Bundestag committees and assigned each speech to one of these categories via zero-shot classification, by embedding those category names with the same fine-tuned model we used to embed the speeches, and assigning each speech to the category with the highest cosine similarity. However, this approach did not meaningfully improve the results: across all six trajectory-based methods and the anchored-axis analysis, we find at most isolated exceptions rather than a consistent crisis effect, mirroring what we saw with BERTopic. Detailed results can be found in the notebook.

6 Trajectories of MPs Who Change Party

Crises act on all parties at once, which is what makes them hard to separate from the era effect. A party switch is the opposite: one person moves while the rest of parliament stays put. It is therefore the cleaner test of whether the embeddings track political position. Of the 20 MPs whose party label changes in the corpus, most cases are the PDS–LINKE merger of 2005–2007 rather than defections; five genuine defections have at least three speeches on either side of the switch and are analysed below, with the merger cases serving as a placebo group.

The measurement follows Section 5.5, with three changes: the mean is subtracted at the MP-year level, the axis runs between the two parties involved in the specific switch rather than between fixed anchors, and switchers are excluded from the centroids they are compared against. The result is expressed in units of the old party's own spread in that year, $z_{i,y} = (p_{i,y} - \overline{p}_{\text{old},y})/\text{sd}(p_{\text{old},y})$, so that $z = 0$ represents an average member of the old party. Because both anchors are re-estimated every year, years a decade apart remain comparable.



Monika Knoche, who left the Grüne for LINKE in 2005, sits inside the ±1 SD band while she belongs to the Grüne, then scores between 2.1 and 3.0 afterwards, above the average LINKE member in three of her four years. Nothing moves beforehand. Martin Hohmann shows the same shape on the CDU/CSU–AfD axis, from 0.98 to 2.79, though thirteen years out of parliament separate the two halves of his series.

To ask whether these movements are unusual rather than merely large, we compute each switcher's cosine distance between a pre- and a post-switch window and rank it against MPs from the same party who lived through the same years and did not switch. Three of the five moved further than 95% of their party colleagues (percentiles 0.95, 0.96 and 0.98). Uwe Kamann at 0.89 is borderline and rests on six speeches either side. Martin Hohmann at 0.50 is not detected by distance at all, yet ends 2.8 standard deviations outside the CDU/CSU on the anchored axis: his windows are fifteen years apart, so every control in that comparison also had a decade and a half to change. Distance and direction fail on different cases, which is why we report both.

The merger group serves as a placebo. Their party label changed in exactly the way a defector's does, and their median percentile is 0.48, against 0.89 for the defections, so the test responds to the defection and not to the relabelling. Three of the twelve nevertheless exceed the 94th percentile, two of them on four speeches, which amounts to roughly one unexplained false positive in twelve. These two sparse false positives point at a general property of the measure. Among MPs who never switched, the fewer speeches behind a window, the further apart the two window vectors appear: the median distance is 0.24 at five or fewer speeches against 0.13 above twenty, correlating +0.25 with $1/\sqrt{n}$. Part of any measured movement is therefore estimation error rather than political change, which is why cells below three speeches are dropped throughout and why the percentile, not the raw distance, is the quantity we interpret. A second limitation is calendar alignment: our windows are years, but MPs defect mid-year, so a switch in July puts speeches from both sides of it into the same cell.

7 Conclusion

Testing the embeddings by various methods showed that fine-tuning modestly sharpens the political signal on most measures we tried, but the results also indicate that other sources of variation are encoded alongside it. Clustering by topic suggests that topic might be the dominant factor driving the embedding space, but the trajectory and event analyses tell a different story: era and government status emerge as the main axes of variation in the raw party-year embeddings, producing an ideologically implausible ordering in naive principal-component trajectories (the AfD–LINKE horseshoe). Once we filter by topic and anchor axes on known reference parties, a crisis-response signal appears only for the 2008 financial crisis. For the other events, the corpus shows no consistent movement in the crisis year, even for Fukushima, where the CDU/CSU is known to have adopted the Grüne's position within months.

We read this as a limitation of what plenary language captures, not as evidence that no realignment occurred. The strongest positive evidence comes from individual party switchers: Knoche and Hohmann move detectably toward their new party's average on an axis anchored on the two parties involved, a movement absent from the PDS–LINKE merger placebo group. Overall, the evidence supports a crisis-response story only for the financial crisis and a change-in-position story only at the individual level, not a broader ideological realignment across the two decades.

Set against classical scaling, this pattern is not surprising. Wordfish and Wordscores place texts on a single latent dimension using word frequencies alone, with word-level weights a researcher can inspect directly to see what drives the scale. Embeddings drop that interpretability, since with 1024 dimensions we cannot read off what a direction means and instead have to validate it indirectly, here done against party labels and CHES scores (Section 4). They also drop the single-dimension assumption, letting several axes (party, era, government status, topic) coexist in the same space, and support tasks Wordfish is not built for, such as measuring one MP's movement relative to their own party (Section 6), at the cost of far more data and compute for a validated ideology signal no stronger than what Wordfish already achieves more cheaply.

Three limitations bear on how far these findings generalise. First, aggregation choices matter: MP-year cells give more stable estimates than MP-month cells but cannot resolve within-year timing, and Section 6 shows measured movement is partly an artefact of speech count under sparse windows. Second, topic and agenda confound position: while some ideological signal is already recoverable from the raw, unfiltered embeddings (Section 4), obtaining an interpretable, orderable position axis for the crisis analysis required explicit topic-filtering and axis-anchoring. Even then, an informative signal emerged only for the financial crisis, while the anchored axes for the other events still partly track government status or party presence rather than policy position. Third, comparability across time is limited by which parties are even present, since the AfD enters only in 2017, the PDS folds into LINKE by 2007, and the FDP drops out of the Bundestag between 2013 and 2017, so party-level comparisons rest on different denominators across periods.

For computational political science, the exercise suggests where embeddings are and are not the right tool. They need no hand-labelled reference texts, unlike Wordscores, and are not committed to a single dimension in advance, unlike Wordfish, and the same pretrained multilingual backbone can in principle be pointed at other legislatures without rebuilding a scaling model. Fine-tuning offers a lightweight lever to steer a general-purpose representation toward a specific target using only weak, indirect supervision. But our results suggest embeddings are currently better suited to exploratory and individual-level tasks, such as topic discovery or detecting a single MP's break from their party, than to precise, validated ideological scaling, where classical, interpretable methods remain the safer default.